

Pranshu Patel

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Summary

2027 Information Technology undergraduate focused on Java backend development with hands-on experience building Spring Boot applications and exposure to AI/LLM-based systems. Experienced in designing REST APIs, handling concurrent workflows, and building reliable backend services.

Education

L.D. College of Engineering (LDCE)

Bachelor of Engineering in Information Technology — CGPA: 8.8/10

July 2023 – June 2027

Ahmedabad, Gujarat

Technical Skills

Languages: Java, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, JDBC

AI & Data: Spring AI, LLM Integration (Groq / Ollama), pgvector, JSQParser

Concepts: Modular Monolith Architecture, JWT Authentication, RBAC, DTO Pattern, Pessimistic Locking, Idempotency, Transaction Management, Pagination & Sorting

Testing & DevOps: JUnit 5, Mockito, Docker, Docker Compose, Git, Maven, Postman, Swagger/OpenAPI, Flyway, Resilience4j

Databases: MySQL, PostgreSQL, Redis

Projects

Ticket Booking System [[GitHub](#)] [[Live Demo](#)] — Spring Boot, PostgreSQL, Redis, Resilience4j, Flyway *March 2026 – June 2026*

- Built a production-style ticket booking backend using a **reserve-pay-confirm** workflow to prevent duplicate seat bookings while keeping database locks out of the payment call.
- Handled abandoned reservations and payment failures with a scheduled reservation reclaimer, Resilience4j retry/circuit breaker, and payment idempotency backed by PostgreSQL and Redis.
- Validated concurrency with an **8-thread race on a single seat**, consistently producing **1 booking and 7 rejections** across 3 runs; the full suite contains **109 tests with 0 failures**.

Secure Engineer Management API [[GitHub](#)] [[Live Demo](#)] — Spring Boot, PostgreSQL, Docker *Nov 2025 – Feb 2026*

- Built a security-focused Spring Boot REST API with **10+ endpoints**, supporting dynamic filtering, pagination, sorting, DTO-based responses, and centralized exception handling.
- Implemented stateless **JWT authentication** and **RBAC** with method-level authorization using **@PreAuthorize**, including custom handling for **401/403** security responses.
- Achieved **90%+ service-layer test coverage** with JUnit 5 and Mockito; integration and controller tests verify unauthorized access, role restrictions, validation, duplicate-resource, and not-found scenarios.

QuerySense — Intelligent SQL Analytics Agent [[GitHub](#)] — Spring AI, PostgreSQL, pgvector, Redis, Docker *July 2026 – Present*

- Building an AI agent that converts natural-language business questions into SQL using Spring AI and LLMs (Llama 3.3 via Groq), with database schema context retrieved through pgvector similarity search.
- Added SQL validation with JSQParser and a dedicated read-only analytics datasource to restrict unsafe or unintended LLM-generated queries.
- Orchestrating Spring Boot, PostgreSQL, Redis, and Ollama services through Docker Compose with health-checked startup dependencies.

Achievements

McKinsey Forward Program — Completed McKinsey & Company's professional development program focused on problem solving and leadership. (Apr 2026 – Jun 2026)

Certifications

Introduction to Generative AI — Google (Badge) | **Java Certification Test** — KnowledgeGate